Ayoub ABBANI

Software Engineer 4 years of experience

+32472701239 | abbaniayoub@gmail.com | Wavre , Belgium | Portfolio | MyGithub | MyLinkedin

Skills

Fluency in English & French (C1)

Core Areas of Expertise:

GUI Qt5/QML, STL & Boost.
Application Embedded Software.
Object-Oriented Programming/Design.
SOLID Principals & Design Patterns.
TDD, BDD, Clean Code & Optimization.
SW Integration, Automation & CI/CD Pipelines.
Agile/Scrum, Waterfall, V-Model.

Technical Skills:

Languages & Frameworks: C/C++17, Python, C#, Flask, JavaScript.
Scripting & Tools: CLI, Shell Scripting, Docker, SSH, VNC, Mobaxterm vSphere.
Build & Testing: CMake, boost.Test.
Databases: SQL, MySQL, MariaDB, MongoDB.
Version Control & CI/CD: Git, Bitbucket, GitHub, Jenkins.
IDEs: Qt Creator/Editor, VS Code.
Environments: Windows, Linux.

Experience

EUROCONTROL & THALES

Brussels, Belgium
Jun. 2023 - Present

Software Engineer | Surveillance Radar Software (Contractor via **Akkodis)**
1 year, 8 months

As a C++ Software Engineer, Contributed to the full lifecycle development and maintenance of the **SASS-C software** within Surveillance Team, Delivering robust algorithms and top level functionality to civil and military Air Navigation Service Providers (ANSPs), R&D and the ATM industry.

Key Responsibilities and Achievements :

- **Re-engineered a Data Visualization Module (VERIF 24.04):** Redesigned the Screeningfile viewer backend, migrated from Java Swing to C++17 and Qt5.15/QML, and implemented additional math computations to process and visualize high resolution radar data.
- **Improved Radar Data Processing (VERIF 24.04.1):** Implemented a backend C++ support for GEOTIFF files, enabling radar data processing, visualization, and reconstruction.
- **Enhanced Trajectory Reconstruction (VERIF 24.10) :** Contributed to an existing reconstruction module by optimizing and tuning Kalman filter algorithms designed for sensor fusion.
- **Implemented Data Management Module (VERIF 24.10.1):** Delivered **EDIS**, a web/desktop module within VERIF 24.10 with **Python**, **Electron** and **Flask**, to manage and integrate Flight/GPS data in VERIF-DB efficiently.
- **Infrastructure & Technical support**: Drafted technical specifications, contributed to CI/CD **Jenkins** pipelines, provided Tier 3 support by investigating incidents via Mantis and delivering prompt feedback to the users.

Tech & Tools: C++17, Qt5/ QML, Python3, Flask, Electron, HTML CSS JS, Shell Scripting, mariaDB5, VNC, Git & Bitbucket, CMake, GDB/GPROF, SonarQube, Docker, Oracle Linux, CI/CD Jenkins, Mantis & Jira.

STELLANTIS

Tangier, Morocco | IDF, France
Jan. 2022 - Jun. 2023

Software Engineer | EE Propulsion Systems (Contractor via **Expleo)**
1 year, 6 months

As a Software Development Engineer, Contributed to the development of in-house software tools for HIL Automated SW Validation.

Key Responsibilities and Achievements:

- Develop new and maintain existing in-house Software using **C++11/Qt5 & Win32 API**.
- Built +20 external runtime library within AutomationDesk using **ASAM XiL API** and **Python3**.
- Deployed automated testing across multiple ECUs & several Powertrain projects on SCALEXIO benches (**HIL & SIL**)
- Maintain comprehensive documentation of test procedures and software versions.
- Collaborated globally with cross-functional teams, including software developers, controls engineers, and validation engineers in an agile setting.

Tech & Tools: Python 3.x, C++11, ASAM XiL API, Qt5, Git/GitHub, Embedded Software, ControlDesk, VEOS, AutomationDesk dSPACE, INCA & MDA, DiagAlyser,CAN & UDS, COM-API.

STELLANTIS

Casablanca, Morocco
Feb. 2021 - Dec .2021

Software Integration & Validation Engineer | Powertrain
10 months

As a Software Integrator & In Vehicle SW Tester in ICE Projects serial life

Key Responsibilities and Achievements:

- Performed SW Integration/upgrade in vehicles ECUs starting from a given reference electronic configuration
- Executed +30 test plans in HIL SCALEXIO Bench's in CAN/OBD/EE Faults/RCW specifications
- Develop CAPL script to perform CAN/LIN Network/Sensors & Actuators test using diagnostic tools and methods
- Provide Root Cause analysis of EE issues and document issue with description readout, measurements, and log files

Tech & Tools: VECTOR CAN/LIN Interface, CANalyzer, CAPL, DiagAlyser, ETAS ECU & BusInterface , INCA/MDA dSPACE: ControlDesk, HIL Bench (SCALEXIO), Python Scripting, UDS/ CAN Network.

Education

Sultan Moulay Salimane University - FST
Engineer's degree in Mechatronics Engineering

Beni Mellal, Morocco
Sep. 2018 - Jun. 2021

ALTEN National Software Hackathon (AMSA): **Won third prize** for developing an ADAS feature with Computer Vision/Machine Learning, C++ & Raspberry Pi4.
Showcase: Prototype / Final Result.
Dec. 2020

Mohammed V University in Rabat - FS
Bachelor of Science - BS in Physics

Rabat, Morocco
Sep. 2015 - Jun. 2018